ModelIt!

Student Activity Pack

Why Do Some Animals Survive Winter?

The Hidden Science of Insulation, Metabolism, and Staying Alive

Grade 04L2 | NGSS 4-LS1-1, 4-LS1-2 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think polar bears stay warm when it is -40 degrees outside?

2 Why do some animals leave (migrate) before winter instead of staying?

3 Draw what you think is inside an arctic animal's fur or skin that keeps it warm.

4 If an animal gets very cold, what do you think its body does to try to warm up?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Temperature | Food Availability | Body Insulation | Metabolic Rate | Survival Chance

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mild Winter, Plenty of Food

Temperature at 60%, Food Availability at 80%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Harsh Winter, Scarce Food

Temperature at 10%, Food Availability at 20%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Harsh Winter, Abundant Food

Temperature at 10%, Food Availability at 80%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Body Size — Larger animals lose heat more slowly because they have less surface area compared to their volume — this is why arctic animals tend to be large

Activity Level — How much the animal moves — more movement generates more heat but also burns more energy

Group Behavior — Whether the animal huddles with others — penguins huddle to share body heat and reduce heat loss

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do arctic foxes, polar bears, and penguins survive in freezing temperatures that would be dangerous for humans in minutes? Use evidence from your simulation to support your answers.

1. **Why do arctic animals tend to have thicker fur and more body fat than animals in warm climates?**

2. **If an animal cannot find enough food in winter, what are its options for survival?**

3. **How is wearing a winter coat similar to an animal having thick fur?**

4. **Why is hibernation a successful survival strategy — what does it do to metabolic rate?**

5. **How did your model help you understand why some animals survive winter easily while others struggle?**

8 STEM Challenge: Design the Ultimate Winter Survival Shelter

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YOUR ENGINEERING MISSION

Design and test a shelter that keeps a cup of warm water from cooling down, using insulation principles from the animal survival model.

THE SCENARIO

A wildlife rescue center needs to build emergency warming shelters for injured animals during blizzards. Your engineering team must design a shelter using available materials that keeps a warm animal (represented by warm water) from losing heat to the cold environment. Use evidence from your model to choose insulating materials.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU TEST AS INSULATION (COTTON BALLS, BUBBLE WRAP, ALUMINUM FOIL, NEWSPAPER, FABRIC)?
2. HOW WILL YOU MEASURE HEAT LOSS OVER TIME (THERMOMETER READINGS EVERY 2 MINUTES)?
3. HOW WILL YOU MAKE SURE YOUR TEST IS FAIR — SAME AMOUNT OF WATER, SAME STARTING TEMPERATURE?
4. CAN YOU COMBINE MATERIALS TO MAKE EVEN BETTER INSULATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Wildlife Biologists and Conservation Scientists study how animals survive in different environments and develop plans to protect endangered species. They earn $55,000–$95,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 An arctic fox has thick fur and a layer of fat under its skin. How do these features help it survive freezing temperatures?

- **A.** They make the fox heavier so it does not blow away in the wind
- **B.** They provide insulation that slows down heat loss, so the fox's body does not have to burn energy as fast to stay warm
- **C.** They help the fox run faster to escape predators in the snow
- **D.** They absorb sunlight and convert it into body heat

2 In the model, Temperature and Food Availability are external components. Why are these external rather than internal?

- **A.** Because they happen outside the animal's body
- **B.** Because they are environmental conditions the animal cannot control, and the other components respond to them
- **C.** Because they are the most important parts of the model
- **D.** Because they never change during winter

3 The model shows that when Temperature drops, Metabolic Rate increases. This is a NEGATIVE relationship. Why does the animal's body burn energy faster in the cold?

- **A.** Because cold temperatures make food taste better, so the animal eats more
- **B.** Because the body must generate more heat to maintain a safe internal temperature when it is cold outside
- **C.** Because cold temperatures make the animal's muscles stronger
- **D.** Because metabolic rate always increases no matter what the temperature is

4 A student runs the model with Temperature at 10% and Food Availability at 20%. Survival Chance drops very low. Which best explains why?

- **A.** The animal is too cold to move, so it cannot find shelter
- **B.** Cold temperatures raise the metabolic rate, but there is not enough food to fuel that high energy demand, so the animal runs out of energy
- **C.** Low food availability makes the temperature feel even colder
- **D.** The animal's insulation disappears when food is scarce

5 **The model shows that Body Insulation has a NEGATIVE relationship with Metabolic Rate. How does better insulation help an animal survive?**

- **A.** Better insulation makes the animal lighter so it uses less energy moving around
- **B.** Better insulation traps body heat inside, so the body does not need to burn as much energy to stay warm, leaving more energy available for survival
- **C.** Better insulation makes food last longer by keeping it frozen
- **D.** Better insulation raises the temperature around the animal